Week 12: Monitoring and Control: Earned Value Management and Reporting

R&D and Project Management in Computer Science

R&D and Project Management in Computer Science (11117BLG002). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Monitoring answers two questions: Is the project where it should be, and where will it end? This week introduces earned value management, which combines scope, schedule and cost in one set of measures, and its use in the reporting of funded projects [1, 2].

Learning outcomes

By the end of the week, students are expected to compute planned value, earned value and actual cost, and to derive schedule and cost variances and performance indices. They should be able to forecast the cost at completion with several methods. Finally, they should be able to choose suitable methods for measuring the progress of research tasks.

Lecture

Baselines and variances

Control requires a baseline: the approved plan of scope, schedule and cost against which performance is measured. Reporting that lists costs spent and tasks started says little about performance, because it does not relate the money spent to the work accomplished. Earned value management was developed to make this relation explicit, and it is described in a standard of the Project Management Institute and in the textbook of Fleming and Koppelman [1, 2].

The three measures

Planned value (PV) is the budgeted cost of the work scheduled up to the status date. Earned value (EV) is the budgeted cost of the work actually performed. Actual cost (AC) is the cost actually incurred for that work. From these, the schedule variance is EV minus PV and the cost variance is EV minus AC. The schedule performance index SPI is EV divided by PV, and the cost performance index CPI is EV divided by AC. An index below one signals a problem. In the example of Figure 12.1, a 12-month project has a budget of 240 thousand euro. At month 6, the

planned value is 96, the earned value 84 and the actual cost 105, so SPI is 0.875 and CPI is 0.8. The project is behind schedule and over budget.

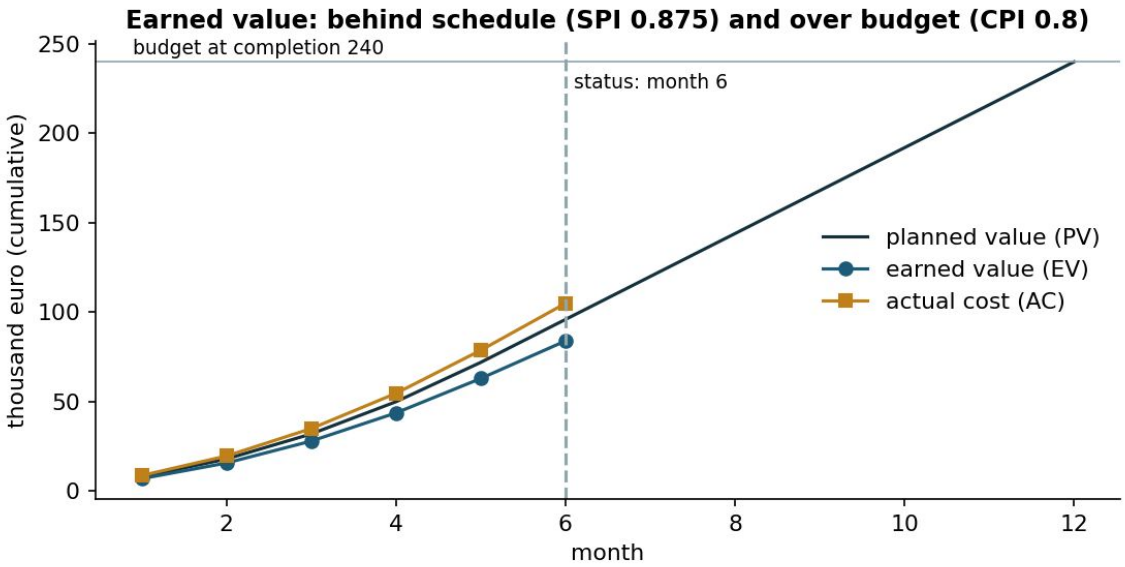


Figure 12.1. Planned value, earned value and actual cost of the example project at month 6 [1].

Forecasting the end

The budget at completion (BAC) is the total planned budget. If the current cost efficiency continues, the estimate at completion is EAC = BAC / CPI, 300 thousand euro in the example. If the overrun was a one-off event and future work will proceed as planned, EAC = AC + (BAC - EV), which gives 261. If both cost and schedule performance influence the remaining work, EAC = AC + (BAC - EV) / (CPI x SPI), which gives about 328. The variance at completion is BAC minus EAC. The to-complete performance index, (BAC - EV) / (BAC - AC), states the cost efficiency that the remaining work would need to finish within budget. A to-complete index well above the current CPI indicates an unrealistic plan. The choice among forecasts is a judgement about the causes of the variance, not a matter of arithmetic.

Measuring progress in research

The weak point of earned value in research is the measurement of EV itself, because progress on uncertain tasks is hard to estimate and percent-complete judgements tend to be optimistic. Fixed rules reduce the problem. The 0/100 rule credits a task only when it is finished, and the 50/50 rule credits half at the start and half at completion. Linking earned value to deliverables and milestones makes it verifiable. Funders require periodic technical and financial reports, and earned value measures, even computed informally, help a project team to see problems before the reports do. The interactive lab computes all measures for three scenarios, and the Colab notebook reproduces the calculations.

> ***Pause and reflect.*** *Which measure of progress would be most honest for the tasks of your project: percent complete, 0/100 or milestones?*

Hands-on components

Interactive lab: Choose a scenario and a status month. The calculator plots planned value, earned value and actual cost, computes variances and indices and compares three forecasts of the cost at completion [1, 2]. Link: https://utkukose.github.io/rd-project-management-NB-lecture/weeks/week-12/lab.html

Colab notebook: It computes earned value measures month by month for the example project and compares forecasts of the cost at completion [1]. Link: https://colab.research.google.com/github/utkukose/rd-project-management-NB-lecture/blob/main/weeks/week-12/NB12_earned_value.ipynb

Reflection prompts

1. Which scenario in the interactive lab resembles a project you know, and which forecast would you have trusted?
2. How would you measure earned value for a task whose outcome is a negative result?
3. What should a periodic report to a funder contain beyond the numbers of earned value management?

Weekly task and submission

What to do. Define the performance measurement baseline of your project and use it once. First, give the planned value by month and the rule for measuring earned value in each work package, such as the 0/100 rule or milestones. Second, assume a status in the middle of the project, for example a delay in one work package. Compute the variances, the indices and a forecast of the cost at completion. Third, write a status report of half a page.

Where to look. Lecture: sections "Baselines and variances", "The three measures", "Forecasting the end" and "Measuring progress in research". Interactive lab: the earned value calculator with its scenarios (tab: Explore). Colab notebook: Section 1 with both exercises.

Example. Suppose that a 10-month project has a budget of 200 thousand euro. At month 4, the planned value is 80, the earned value 70 and the actual cost 75, all in thousand euro. Then SPI = 70 / 80 = 0.875 and CPI = 70 / 75, about 0.93. If this cost efficiency continues, the estimate at completion is 200 / (70 / 75), about 214 thousand euro.

What to submit. The baseline, the status report, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Project Management Institute (2019). The Standard for Earned Value Management. Project Management Institute.

[2] Fleming, Q. W., & Koppelman, J. M. (2010). Earned Value Project Management (4th ed.). Project Management Institute.